A Self-Supervised Transformer With Feature Fusion for SAR Image Semantic Segmentation in Marine Aquaculture Monitoring

Jianchao Fan, *Member, IEEE*, Jianlin Zhou, Xinzhe Wang, and Jun Wang, *Life Fellow, IEEE*

***Abstract*— The rapid development of the marine aquaculture industry has brought about a series of environmental problems that need to be monitored and planned. There is abundant marine aquaculture data obtained through synthetic aperture radar (SAR) remote sensing over a long period. With a large amount of unlabeled data, self-supervised learning can describe the feature representation of targets. However, when self-supervised learning meets big data, it often leads to semantic information loss, such as interclass misjudgment and intraclass discontinuity. To address this issue, this article proposes a self-supervised transformer with feature fusion (STFF) for the semantic segmentation of SAR images in marine aquaculture monitoring. STFF consists mainly of a self-attention encoding module with a hybrid loss function and a semantic segmentation decoding module with feature fusion. For encoding, the transformer is pretrained via self-supervised learning based on a hybrid loss function to enrich local, global, and edge information for dealing with semantic information loss and data imbalance in whole-scene SAR images. For decoding, the features extracted from transformer blocks are fused to enhance semantic characteristics, improve the intraclass continuity of segmentation, and reduce the occurrence of interclass misjudgment. The superiority of the proposed method to state-of-the-art algorithms is demonstrated via experimentation on GaoFen-3 and Radarsat-2 SAR datasets. The code has been available at https://github.com/fjc1575/Marine-Aquaculture/tree/main/STFF-code for the sake of reproducibility.**

***Index Terms*— Marine aquaculture, self-supervised learning, semantic segmentation, synthetic aperture radar (SAR) images, transformer-based model.**

Manuscript received 5 July 2023; revised 19 August 2023 and 25 September 2023; accepted 29 September 2023. Date of publication 4 October 2023; date of current version 13 October 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42076184, Grant 41876109, and Grant 41706195; in part by the National Key Research and Development Program of China under Grant 2021YFC2801000; in part by the National High Resolution Special Research under Grant 41-Y30F07-9001-20/22; in part by the Fundamental Research Funds for the Central Universities under Grant DUT23RC(3)050; and in part by the Dalian High Level Talent Innovation Support Plan under Grant 2021RD04. *(Corresponding authors: Jun Wang; Xinzhe Wang.)*

Jianchao Fan is with the School of Control Science and Engineering, Dalian University of Technology, Dalian 116023, China (e-mail: fjchao@dlut.edu.cn).

Jianlin Zhou is with the School of Automation, Central South University, Changsha 410083, China (e-mail: 234601069@csu.edu.cn).

Xinzhe Wang is with the School of Information Science and Engineering, Dalian Polytechnic University, Dalian 116034, China (e-mail: wxzagm@dlpu.edu.cn).

Jun Wang is with the Department of Computer Science and the School of Data Science, City University of Hong Kong, Kowloon, Hong Kong (e-mail: jwang.cs@cityu.edu.hk).

Digital Object Identifier 10.1109/TGRS.2023.3321595

I. Introduction

MARINE aquaculture, also known as sea farming or mariculture, is the cultivation of marine organisms, such as fish, shellfish, and seaweed in the ocean or other saltwater environments. It has become increasingly popular to meet the growing demand for seafood, reduce pressure on wild fish populations, and promote sustainable food security [1]. Although it brings enormous economic benefits, it also results in pollution in marine ecological environments [2]. Monitoring marine aquaculture activity is essential for the macromanagement of marine resources [3]. Synthetic aperture radar (SAR) image is one of the primary data sources for monitoring marine aquaculture activities. It can be observed that marine aquaculture is widely distributed in coastal areas, and monitoring based on remote sensing can obtain a large amount of unlabeled data.

Convolutional neural networks (CNNs) have succeeded in different applications of remote sensing images [4]. Particularly, fully convolutional networks (FCNs) play an important role in pixel-to-pixel semantic segmentation [5]. With the development of deep learning, CNN is usually adopted as the backbone network, and various improvements are made to improve its performance. For example, operations such as atrous convolution [6], pyramid pooling [7], residual connection [8], and attention mechanism [9] can expand the field of perception, extract multiscale features, select specific inputs, etc., and thus improve the accuracy and efficiency of semantic segmentation. In addition, there are some special structures of semantic segmentation, such as FCN [5], encoding-decoding networks [6], [10], [11], and vision transformers [12], [13]. These methods have achieved satisfactory results in different scenarios.

For marine aquaculture monitoring, the multiscaled attention U-net with dilated convolution and offset convolution (MDOAU-Net) builds on the attention U-net with different convolution blocks to enhance the accuracy and improve the robustness against noises in SAR images [14]. Nevertheless, supervised learning methods entail much label information in marine aquaculture segmentation [15], [16], [17], and unsupervised learning methods seem to alleviate the label demand. For example, a multikernel fuzzy C-means algorithm leverages a neurodynamic optimization method to improve the marine aquaculture segmentation quality of SAR images [18].

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

But unsupervised learning methods are weak in terms of generalization capability [19]. Recently, an incremental double unsupervised deep learning method has been proposed to enhance generalization capability with marine aquaculture segmentation [20]. However, its effectiveness is limited because the whole-scene remote-sensing images contain multiple targets.

As the amount of data increases, self-supervised learning becomes a solution method to balance labeling cost and generalization performance in complex environments. In self-supervised learning, an upstream network is trained with a large amount of unlabeled data, and a downstream network is fine-tuned with a small number of labels [21]. Self-supervised learning is roughly divided into two categories: generative learning and contrastive learning [22]. The generative learning model usually uses autoencoders [23] or generative adversarial networks (GANs) [24]. The original information as input is mapped to the feature space and then restored to the original input information through the generator. However, pixel-level generation is computationally expensive and may not be necessary for representation learning [25]. The contrastive learning has achieved great success [26]. It typically narrows positive sample distance and separates negative samples to obtain better feature representations. SimCLR [25], MoCo [27], and SwAV [28] have presented overwhelming performances on various visual benchmarks. However, the aforementioned self-supervised learning methods based on convolution focus mainly on local features with constrained receptive fields.

Consisting of self-attentive blocks, a transformer enables images to embed into a vector to obtain global information against disturbances [29]. The self-supervised transformers explore explicit information without labels [30]. The mask-images and classification-similarity loss function are combined as pretext tasks to acquire visual semantics [31]. The transformer's ability to process global features and large amounts of data has been gradually developed in remote sensing. Several transformer models obtain more spatial global information [32] and expand spatial perception fields [33]. With the expansion of space, large-sized remote-sensing images often suffer from the small interclass variance, resulting in misjudgment of targets. The multiscale fusion [34], class-guided classification [35], and salient feature guidance [36] based on a transformer enlarge the variances and obtain fine-grained information. The SAR remote-sensing images from marine aquaculture suffer from the problems of interclass misjudgment and intraclass discontinuity. Although the above methods alleviate this problem from multiscale fusion and other aspects, it still needs to consider the semantic information of marine aquaculture.

Targeting the above problems, this article presents a self-supervised transformer with feature fusion (STFF) for the semantic segmentation of SAR images in marine aquaculture monitoring. STFF consists of a transformer encoding module and a semantic segmentation decoding with a feature fusion module. To obtain multitype features and reduce label requirements, STFF utilizes a transformer based on self-distillation with a global contrast-learning (CL) loss, a local masked-image loss, and an edge loss function. To acquire more semantic information, STFF extracts high-level features from the last three transformer blocks. Then, they are diffused to merge with low-level features from the first few transformer blocks. Eventually, high-level and low-level features are fused and fed into output layers to obtain semantic segmentation results.

The main contributions of this article are summarized as follows.

1) An STFF is proposed. It can fully utilize a large amount of unlabeled remote sensing data of marine aquaculture to acquire long-term dependence features. In addition, STFF can process whole-scene SAR images, including multiple targets, such as aquaculture areas, sea, and land.
2) A hybrid loss function is designed in self-supervised learning to obtain various types of adequate information, including CL loss, masked-image loss, and edge loss function. Based on this function, global information, local semantics, and edge information of marine aquaculture can be explored synchronously to improve the accuracy of SAR image semantic segmentation.
3) A feature fusion module is proposed to enhance the semantic information of deep layers and retain the shape information of shallow layers. The shape information retainment has utilized the features extracted from different layers of the encoding module to maintain target edges and reduce the occurrence of interclass misjudgments. The semantic information enhancement is focused on deep semantic continuity by fusing high-level features to improve the intraclass continuity of segmentation images.

The rest of this article is organized as follows. The related work is briefly reviewed in Section II. The proposed STFF for SAR image semantic segmentation is presented in Section III. Experimental results are discussed in Section IV. Finally, the conclusions are provided in Section V.

II. Related Work

The attention mechanism assigns different attention weights to different contents in the process of information processing. Recently, there have been many attention mechanisms applied to visual models [37], [38], [39]. The transformer is a new type of neural network, mainly using the self-attention mechanism to extract features, as shown in Fig. 1. The transformer model is divided into input embedding and encoding modules, as shown in Fig. 1(a). It consists of N modules, and the core of each module is the multihead attention module, as shown in Fig. 1(b). The multihead attention module is a combination of multiple self-attention. The input X is converted into the query vector Q, the key vector K, and the value vector V

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where

$$\begin{cases} Q = W^q X \\ K = W^k X \\ V = W^v X \end{cases}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

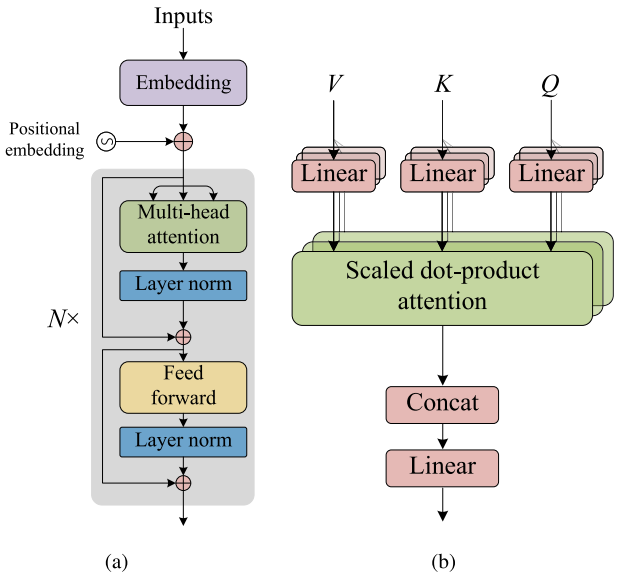


Fig. 1. Transformer model. (a) Structure of original transformer encoder and (b) multihead attention module.

W^q, W^k, and W^v are optimization parameters, respectively. QK^T gets the attention weight. d_k is the dimension of K to ensure the gradient stability of softmax. V gets the weighted sum while remaining consistent with the input.

The multihead attention module allows the model to focus on different aspects of information from multiple subspaces. However, the multihead attention module lacks position information, so position embedding is introduced

$$\mathrm{PE}(\mathrm{pos}, 2i) = \sin\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\right) \tag{2}$$

$$\mathrm{PE}(\mathrm{pos}, 2i+1) = \cos\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\right) \tag{3}$$

where pos denotes the position of the patch in the image, i denotes the dimension index of the patch, and d_{model} denotes the total dimension of the patch. The position embedding is added to the input embedding to capture the position in the sequence. The transformer is mainly composed of the self-attention blocks. The high-layer attention of the transformer coder tries to capture long-range dependences. In contrast, low-layer attention tries to find local dependences [40]. To fully use the transformer's strong representation capability, researchers have made many attempts for feature decoders. For example, the encoder and decoder by skip connection obtain information about the high and low layers simultaneously [41]. The multiscale feature fusion expands the receptive field of convolution operation in the decoder [42].

III. Proposed Model

From the self-supervision paradigm perspective, STFF consists of two parts, as shown in Fig. 2. The first part is to pretrain a transformer encoder through self-distillation with the hybrid loss function. The second part is to fine-tune the semantic segmentation decoder through the feature fusion module.

A. Self-Supervised Learning of STFF

Self-supervised learning utilizes pretext tasks to obtain potential features of unlabeled data. Most self-supervised learning algorithms focus on learning global information and perform well in classification tasks. In aquaculture area segmentation, local features are needed to restore fine-grained information about targets. In order to extract the local information, the segmentation-oriented image mask task is added. The central idea is to find invariant information between two views (masks) by random data augment (image masks). In addition, an edge task is designed to enhance the margin of the aquaculture area. The core idea is to strengthen the edge semantics of acquired features based on an unsupervised segmentation algorithm.

During the training process of the encoder, two structurally identical transformer encoders, called the teacher network $f_{\theta_t}(\cdot)$ and the student network $f_{\theta_s}(\cdot)$, will be constructed based on self-distillation learning, a kind of self-supervised learning method. The encoder optimization is performed through a global CL loss, a local masked-image loss, and an edge loss function, as shown in Fig. 3. The self-supervised learning of the STFF algorithm is as shown in Algorithm 1.

1) CL and Masked-Image Loss Functions: Specifically, the original image x is subjected to two random and independent data augmentations to obtain two different views u and υ [30]. The two different views obtain their masked views $\hat{u}$ and $\hat{\upsilon}$ from embedding and masking [31]. After that, the deep features $f_{\theta_t}(u, \upsilon)$ and $f_{\theta_s}(\hat{u}, \hat{\upsilon})$ are obtained from the transformer encoder. The different full connection layers, called CL and mask image (MI), respectively, are added to handle the CL loss and masked-image loss.

The contrast loss function is the key to CL, and the main idea is to find similarities under different views. The results obtained from the teacher network f_{θ_t} are input into the softmax function

$$\mathrm{SF}_t(u_c, \upsilon_c)^{(i)} = \frac{\exp\left(\left(f_{\theta_t}(u_c, \upsilon_c)^{(i)} - C\right)/\tau_t\right)}{\sum_{k=1}^{K} \exp\left(\left(f_{\theta_t}(u_c, \upsilon_c)^{(k)} - C\right)/\tau_t\right)} \tag{4}$$

where i and k are the network output probability distributions over K dimensions. C is centered with a mean computed over the batch of the teacher network output to avoid collapse. τ_t denotes a temperature parameter with the teacher network. The results of the student network f_{θ_s} are also input into softmax function

$$\mathrm{SF}_s\left(\hat{u}_c, \hat{\upsilon}_c\right)^{(i)} = \frac{\exp\left(f_{\theta_s}\left(\hat{u}_c, \hat{\upsilon}_c\right)^{(i)}/\tau_s\right)}{\sum_{k=1}^{K} \exp\left(f_{\theta_s}\left(\hat{u}_c, \hat{\upsilon}_c\right)^{(k)}/\tau_s\right)} \tag{5}$$

where τ_s denotes a temperature parameter with the student network. The cross entropy loss is used to expect the output of the teacher and student networks to be matched to optimize student network parameters

$$\begin{aligned} L_{\mathrm{CLS}} &= -\mathrm{SF}_t(u_c, \upsilon_c) \log \mathrm{SF}_s\left(\hat{u}_c, \hat{\upsilon}_c\right) \\ &= -\frac{1}{2}\left(t_{c_1} \log s_{c_2} + t_{c_2} \log s_{c_1}\right) \end{aligned} \tag{6}$$

where L_{CLS} is the CL loss function. Specifically, t_{c_1} is the teacher network separated CL features $\mathrm{SF}_t(u_c)$ of view u_c. t_{c_2} is the teacher network separated CL features $\mathrm{SF}_t(\upsilon_c)$ of view υ_c. s_{c_1} is the student network separated CL features $\mathrm{SF}_s(\hat{u}_c)$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

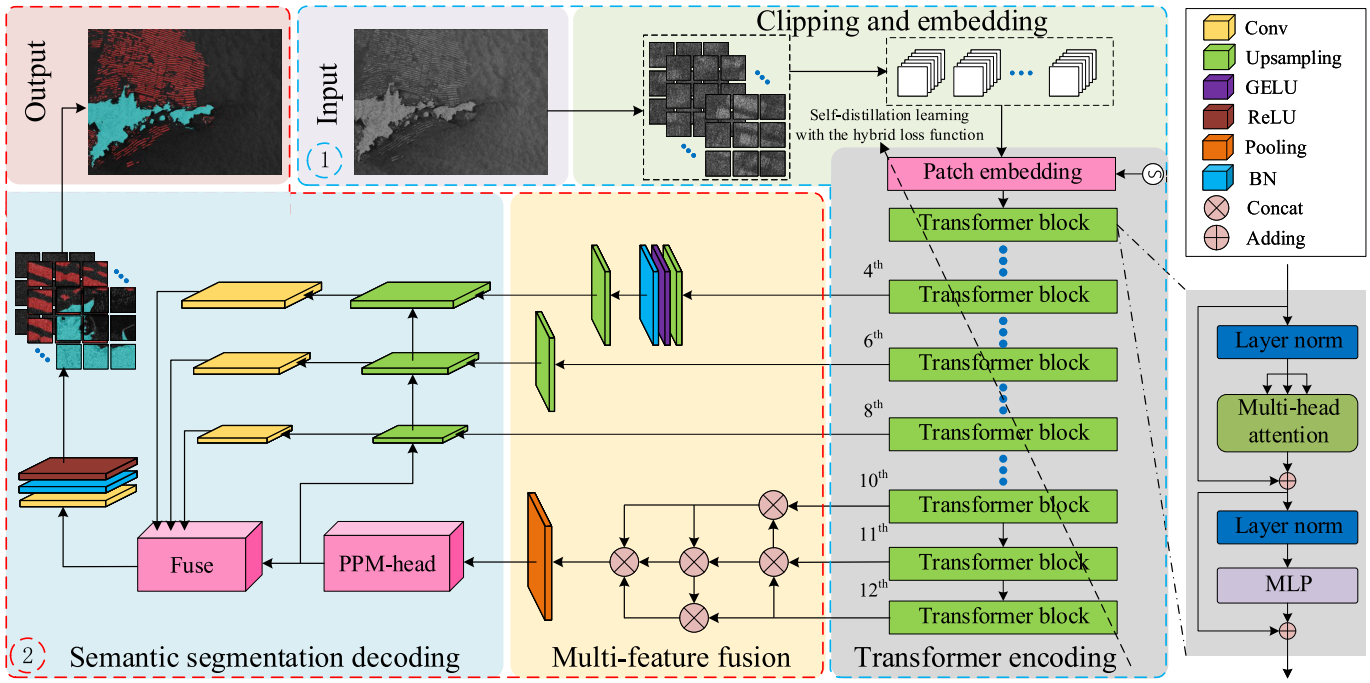


Fig. 2. STFF structure including the remote-sensing image input, the transformer encoder consisting of 12 transformer blocks, the semantic segmentation decoder with feature fusion. The self-distillation learning with the hybrid loss function is used to obtain the transformer encoding parameters.

of masked view $\hat{u}_c$. s_{c_2} is the student network separated CL features $\mathrm{SF}_s(\hat{\upsilon}_c)$ of masked view $\hat{\upsilon}_c$.

The masked-image loss function is expected to reconstruct the features from the teacher network features with no mask, i.e., part of the MI does not affect the overall interpretation of the image. Like the CL fully connected layer, the results need to enter the softmax function, where the hyperparameters τ_t' and τ_s' and center C' are independent of each other even though they have the same dimensions K

$$\mathrm{SF}_t'(u_m, \upsilon_m)^{(i)} = \frac{\exp\left(\left(f_{\theta_t}(u_m, \upsilon_m)^{(i)} - C'\right)/\tau_t'\right)}{\sum_{k=1}^{K}\exp\left(\left(f_{\theta_t}(u_m, \upsilon_m)^{(k)} - C'\right)/\tau_t'\right)} \tag{7}$$

$$\mathrm{SF}_s'\left(\hat{u}_m, \hat{\upsilon}_m\right)^{(i)} = \frac{\exp\left(f_{\theta_s}\left(\hat{u}_m, \hat{\upsilon}_m\right)^{(i)}/\tau_s'\right)}{\sum_{k=1}^{K}\exp\left(f_{\theta_s}\left(\hat{u}_m, \hat{\upsilon}_m\right)^{(k)}/\tau_s'\right)}. \tag{8}$$

Subsequently, the loss function L_{MIS} is defined according to the idea of masked image

$$\begin{aligned} L_{\mathrm{MIS}} &= -m_{u,\upsilon}\mathrm{SF}_t'(u_m, \upsilon_m)\log \mathrm{SF}_s'\left(\hat{u}_m, \hat{\upsilon}_m\right) \\ &= -\frac{1}{2}\left(\frac{\left(m_u t_{m_1}\log s_{m_1}\right)}{m_u} + \frac{\left(m_\upsilon t_{m_2}\log s_{m_2}\right)}{m_\upsilon}\right) \end{aligned} \tag{9}$$

where t_{m_1} is the teacher network separated masked image features $\mathrm{SF}_t'(u_m)$ of view u_m. t_{m_2} is the teacher network separated masked image features $\mathrm{SF}_t'(\upsilon_m)$ of view υ_m. s_{m_1} is the student network separated masked image features $\mathrm{SF}_s'(\hat{u}_m)$ of masked view $\hat{u}_m$. s_{m_2} is the student network separated masked image features $\mathrm{SF}_s'(\hat{\upsilon}_m)$ of masked view $\hat{\upsilon}_m$. $m_{u,\upsilon}$ is the mask encoding.

2) Edge Loss Function: Through self-distillation learning, the explicit feature can be obtained [30]. In order to take advantage of this feature, STFF designs the edge loss. Edge loss expectation enhances the extraction of the aquaculture target edge. The specific implementation is as follows. First, the original image is subjected to an unsupervised k-means algorithm to obtain the binary classification map. In order to match the size of the feature map, the binary classification map is divided into 14×14 sizes. The dominance ratio is judged in each small block, i.e., to determine whether there are more black or white areas in the small block, and the color that is more is substituted for the whole block to obtain the edge information. The feature map is subjected to a similar operation, and the feature edge map is obtained similarly. Finally, the edge loss function is designed

$$L_{\mathrm{EL}} = \sum_{n=1}^{N}(p_i - e_i)^2 \cdot \mathrm{mean}() \tag{10}$$

where N is the batch size, e_i is the edge image consisted of 0 and 1, and p_i is the edge of explicit feature map.

3) Self-Supervised Learning Procedure: Based on the CL loss, the masked-image loss, and the edge loss function, STFF adopts self-distillation to learn the encoder parameters. The student network parameters θ_s are optimized by the hybrid loss function L_{all}, which captures the deep semantics of different views in the category and the visual semantics of mask patch reconstruction and edge enhancement

$$L_{\mathrm{all}} = L_{\mathrm{CLS}} + L_{\mathrm{MIS}} + \alpha L_{\mathrm{EL}}. \tag{11}$$

The teacher network parameters θ_t are updated with an exponentially moving average based on the parameters of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

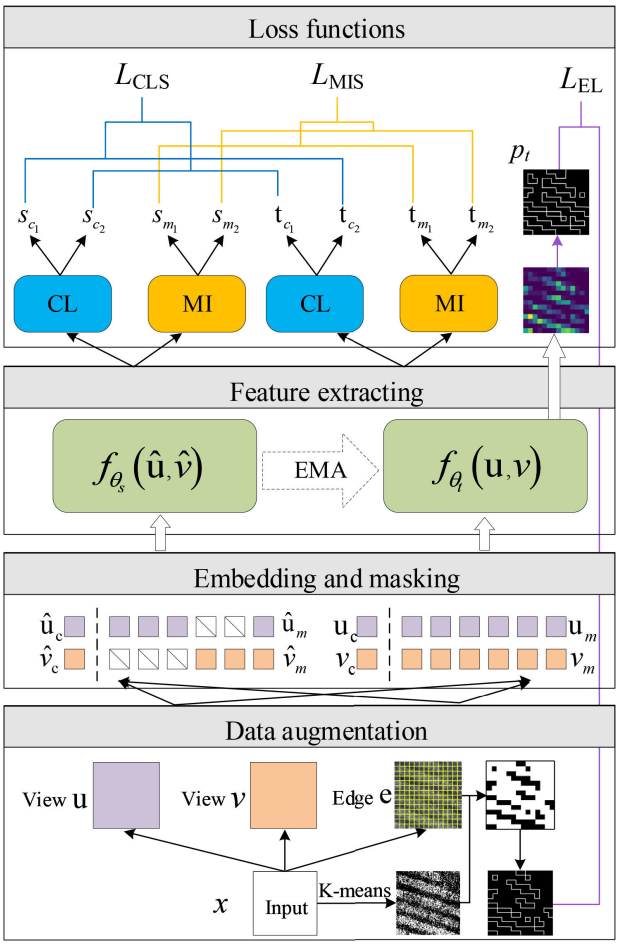


Fig. 3. Self-supervised learning diagram of STFF with global contrast loss L_{CLS}, local masked-image loss L_{MIS}, and edge loss function L_{EL}. MI is the masked-image feature extraction heads, CL is the CL feature extraction heads, and EMA is the weights update strategy of the exponential moving average.

student network θ_s. After the experiment, the hyperparameter α is set to 0.1

$$\theta_t \leftarrow \lambda\theta_t + (1-\lambda)\theta_s \tag{12}$$

where λ is the momentum parameter from 0.996 to 1. Finally, the STFF model uses the trained teacher network as the transformer encoder.

B. Feature Enhancement and Fusion

The feature fusion module of STFF is to emphasize the depth semantics of features based on fusing shallow features and depth features. The shallow features contain the shape information to improve the edge extraction, and the depth features contain the semantic information to improve the misjudgment and discontinuity. However, the shallow features also include speckle noises. To avoid the noises, the depth semantics of target features are important and need to be emphasized and diffused to the shallow features. Finally, the feature fusion module fuses the diffused shallow features with the semantically enhanced depth features. So the feature fusion module is composed of two parts: multilayer feature fusion and deep semantic enhancement.

Algorithm 1 STFF Learning Algorithm

Require: $x, f_{\theta_s}, f_{\theta_t}, \tau, k, \lambda, \alpha$

for x **in loader:**

$u, \upsilon = Augment(x), Augment(x)$
$x_k = Kmeans(x)$
$e_x = DetectEdge(x_k)$

$u_c, \upsilon_c, u_m, \upsilon_m = InputEmbedding(u, \upsilon)$

$m_u, \hat{u}_c, \hat{u}_m = RandomMask(u_c, u_m)$
$m_\upsilon, \hat{\upsilon}_c, \hat{\upsilon}_m = RandomMask(\upsilon_c, \upsilon_m)$

$t_{c_1}, t_{c_2}, s_{c_1}, s_{c_2} = H(f_{\theta_t}, f_{\theta_s}, \tau_t, \tau_s, C)$
$t_{m_1}, t_{m_2}, s_{m_1}, s_{m_2} = H(f_{\theta_t}, f_{\theta_s}, \tau'_t, \tau'_s, C')$
$f_e = Kmeans(f_{\theta_t}(u))$
$p_t = DetectEdge(f_e)$

$$Loss_{\text{CLS}} = -\tfrac{1}{2}(t_{c_1}\log s_{c_2} + t_{c_2}\log s_{c_1}) \tag{6}$$
$$Loss_{\text{MIS}} = -\tfrac{1}{2}\left(\frac{m_u t_{m_1}\log s_{m_1}}{m_u} + \frac{m_\upsilon t_{m_2}\log s_{m_2}}{m_\upsilon}\right) \tag{9}$$
$$Loss_{\text{EL}} = (p_t - e_x)^2.mean \tag{10}$$

$loss = Loss_{\text{CLS}} + Loss_{\text{MIS}} + \alpha Loss_{\text{EL}}$
$loss.backward()$

update θ_s
$\theta_t = \theta_t + (1-\lambda)\theta_s$
$C = k * C + (1-k) * cat([t_{c_1}, t_{c_2}]).mean$
$C' = k' * C' + (1-k') * cat([t_{m_1}, t_{m_2}]).mean$
end

def $H(f_{\theta_t}, f_{\theta_s}, \tau_t, \tau_s, C)$:
$f_{\theta_t} = f_{\theta_t}.detach()$
$s = softmax(f_{\theta_s}/\tau_s, dim = 1)$
$t = softmax((f_{\theta_t} - C)/\tau_t, dim = 1)$
return t, s

1) Deep Semantic Enhancement: The deep semantic enhancement depends on the high level of the transformer encoder, and STFF makes the high-level semantic information contain more transformer encoder layers through feature enhancement. Unlike previous works based on deep convolutional features, deep semantic augmentation is based on deep transformer features. Selecting more deep features of the encoder network can be adapted to semantic segmentation. Specifically, different layer features E_m are obtained from the encoder of STFF to contain richer semantic information. The deep semantic enhancement has contained the features $E_m, m \geq 10$, utilizing the fully connected and concat operation

$$C_{10} = \text{FC}(\text{concat}(E_{10}, \text{FC}(\text{concat}(E_{11}, E_{12})))) \tag{13}$$
$$C_{11} = \text{FC}(\text{concat}(C_{10}, \text{FC}(\text{concat}(E_{11}, E_{12})))) \tag{14}$$
$$C_{12} = \text{FC}(\text{concat}(C_{11}, E_{12})) \tag{15}$$
$$F_{\text{high}} = \text{FC}(\text{concat}(C_{10}, C_{11}, C_{12})) \tag{16}$$

where FC$(\cdot)$ is a fully connected convolution.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

The high-level features after the enhancement are taken as the final high-level features F_{high} and spread to the low-level features after the pyramid pooling module (PPM) [43]. The PPM utilized different pooling sizes to obtain the extensive high-level semantic information in the previous step

$$F_{\text{PPM}} = \text{concat}\left(\xi\left(\text{Pool}\left(F_{\text{high}}, \text{size} = 1, 2, 3, 6\right)\right)\right) \tag{17}$$

where $\xi(\cdot)$ is a bilinear upsampling function.

2) Multilayer Feature Fusion: STFF uses the convolution decoder, which considers semantic segmentation of visual tasks and a combination of multilevel features. The transformer encoder has powerful feature representation ability. The low-level transformer can extract the underlying features of the image, such as color and texture, and the high-level transformer can extract the semantic features of the image. Therefore, STFF contains high-level and low-level information from the transformer encoder.

The multilayer feature fusion is diffused to the low-level features to reduce the noise of the low-level features. First, low-level features are upsampled, respectively, and the times of upsampling differ according to the size of low-level features. Because lower level information contains more noise and local information, the underlying features of the target can be restored by further expanding the feature map. Specifically, different layer features E_m are obtained from the encoder of STFF to contain low-level and high-level information. The multilayer feature fusion contains the features E_m, $m = \{4, 6, 8\}$ and high-level features F_{PPM}, utilizing conventional convolution

$$F_4 = \xi'\left(\text{BN}\left(\sigma\left(\xi'(E_4)\right)\right)\right) \tag{18}$$

$$F_6 = \xi'(E_6) \tag{19}$$

where σ is the activate function, ξ' is the upsampling of transposed convolution, and BN is the batch normalization. Unlike the previous bilinear upsampling, the transformer deals with feature vectors and can be adapted to downstream visual tasks through transposed convolution. After obtaining the features of different layers and sizes, the decoder needs to diffuse the deep feature to the lower layer and fuse the diffused low layer feature with the deep feature

$$A_8 = \phi(\xi(\phi(F_{\text{PPM}})) + E_8) \tag{20}$$

$$A_6 = \phi(\xi(A_8) + F_6) \tag{21}$$

$$A_4 = \phi(\xi(A_6) + F_4) \tag{22}$$

$$Y_{\text{out}} = \sigma(\text{BN}(\phi(\text{concat}(A_4, A_6, A_8, \phi(F_{\text{PPM}}))))) \tag{23}$$

where ϕ is a convolution with a 3×3 filter. The operation of high-level semantic fusion, low-level information diffusion, and multilayer information fusion can effectively enhance the discrimination ability to reduce misjudgment and improve the continuity segmentation ability of the network to the aquaculture region.

IV. Experiments

The experiments include three parts: 1) the applicability of different datasets; 2) the effectiveness of the feature fusion module and the edge loss function; and 3) the generalizability of downstream task fine-tuning. The evaluation criteria include overall accuracy (OA), and recall (R), precision (P), kappa, and mean intersection over union (mIoU). Several methods are compared with the performance of different supervised methods (i.e., U-Net [10], Deeplabv3+ [6], MDOAU-Net [14], ViT [12]) and self-supervised methods (i.e., SimCLR [25], IBot [31]).

The hyperparameters of interest in STFF are obtained through grid search experiments. The transformer encoder has 12 blocks, the input embedding patch size is 16×16, and the embedding channel is 384. The data augment part resizes the input image to 224×224. The experimental results show that the hyperparameters k and k' are 0.9, $\tau_t = 0.04$, $\tau_t' = 0.07$, $\tau_s = \tau_s' = 0.1$, and $\lambda = 0.996$ referred to [30] and [31] achieve the best results.

A. Datasets

Raft aquaculture is one of the main marine aquaculture types, consisting of floats and ropes. It suspends seaweed and shellfish on the sling below the sea's surface. SAR imaging is more affected by floating balls, and scattered floating balls may cause missing judgment for extracting aquaculture areas. Therefore, mIoU and *kappa* are more critical in evaluating the semantic continuity of marine aquaculture area extraction.

The experiments are based on two different SAR satellite images, GaoFen-3 (GF-3) and Radarsat-2. As shown in Fig. 4, GF-3 datasets select four different satellite image areas, A, B, C, and D, according to the proportions area of aquaculture, land, and seawater. The proportion of aquaculture area and land is balanced in areas A and D, as shown in Fig. 4(a) and (b). The proportion of sea is relatively large in area B, as shown in Fig. 4(b). The proportion of land and sea is even in area C, as shown in Fig. 4(c). Similarly, in Radarsat-2 datasets, image areas E–H of different areas are selected, as shown in Fig. 5.

GF-3 and Radarsat-2 SAR images in Changhai, China, are cropped to 512×512. In the GF-3 experiment, more than 50 000 unlabeled and 300 labeled images are utilized for self-supervised training and fine-tuning, respectively. And 1220 images are used in the test. In the Radarsat-2 experiment, more than 10 000 unlabeled and 135 labeled images are adopted for self-supervised training and fine-tuning, respectively. And 540 images are employed in the test.

B. Results on GF-3 Marine Aquaculture Datasets

In the GF-3 datasets, Fig. 6 displays the experimental results of whole-scene remote-sensing segmentation images. Table I tabulates the corresponding accuracies. It suggests that the proposed one achieves the best performance on the GF-3 datasets. Fig. 6(a) shows that U-Net has poor continuity in the land region. Fig. 6(b) shows that Deeplabv3+ has more sporadic misclassification in the seawater region. Fig. 6(c) shows that MDOAU-Net is weak at recognition of land. Fig. 6(d) shows that SimCLR and ViT display a rectangle phenomenon, and IBot misclassifies the aquaculture areas in the upper right corner as the land region. In addition, Fig. 7 shows the four regions of interest in detail on the GF-3 datasets, the locations

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

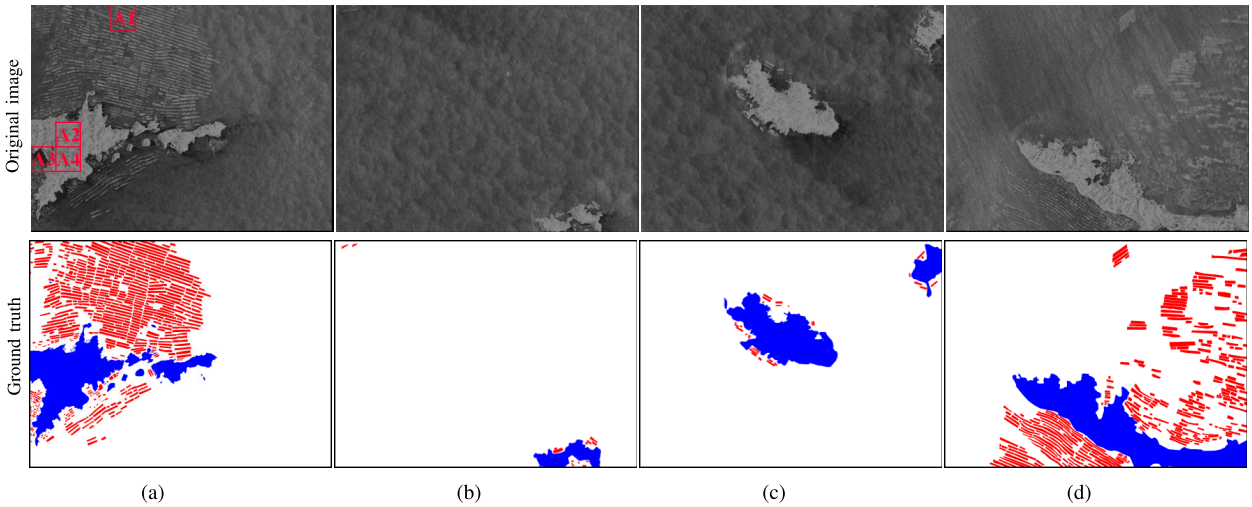


Fig. 4. Original images and ground truth of GF-3 aquaculture datasets. (a) Area A. (b) Area B. (c) Area C. (d) Area D.

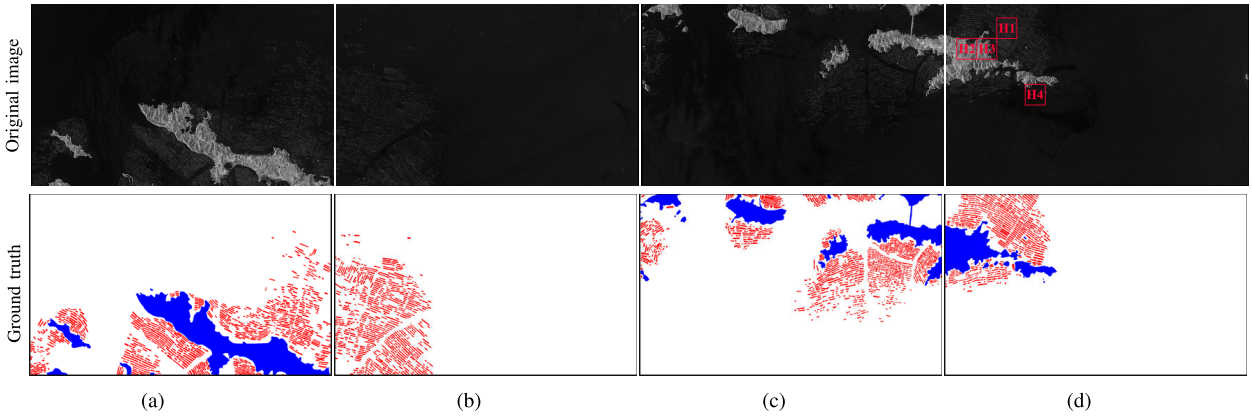


Fig. 5. Original images and ground truth of Radarsat-2 aquaculture datasets. (a) Area E. (b) Area F. (c) Area G. (d) Area H.

of which are boxed in red in Fig. 4. Due to the complexity of whole remote-sensing images, the selected areas are displayed according to the classification type. In the case of one class, land area A2 is selected; in the case of two classes, aquaculture seawater area A1 and land seawater area A4 are selected; in the case of three classes, aquaculture area, land, and seawater area A3 are selected.

CNN-based methods are easily interfered with by speckle noises in SAR images due to the limitation of the receptive field and semantic information. As shown in Fig. 7(d), Deeplabv3+ occurs adhesion in the aquaculture areas. As shown in Fig. 7(c), some misjudgments occur with U-Net in the marine aquaculture areas. The above networks do not learn the semantic features of the aquaculture area, mainly manifested by unreasonable segmentation. The continuity is disturbed, such as mixed land in aquaculture areas and sporadic aquaculture areas in distant offshore areas, because of a lack of semantic learning ability. As shown in Fig. 7(e), the phenomena of the two above methods are balanced by MDOAU-Net, designed to avoid interference of land and speckle noises and to reduce misjudgment by enlarging the receptive field and obtaining fine-grained information. At the same time, the misjudgment of aquaculture areas in the distant offshore area is avoided. MDOAU-Net was initially designed to avoid land interference, forcing the land to be ignored and resulting in poor land segmentation on this dataset. However, expanding the receptive field, i.e., multiscale information, is based on convolution and upsampling and does not consider deep semantic features. Transformer-based methods fundamentally alleviate this problem. ViT learns global features, including semantic information, through the transformer model. As shown in Fig. 7(f), the results consistent with semantic continuity are extracted based on deep semantic features. However, the rectangular phenomenon is severe due to the method based on image patches. As shown in Fig. 7(i), the proposed one avoids the lack of aquaculture area semantics through deep semantic enhancement and the rectangle phenomenon caused by the lack of information through a multilayer feature fusion module.

For self-supervised learning, as shown in Fig. 7(g), SimCLR has a severe rectangle phenomenon caused by insufficient features during upsampling due to the limitation of computational amount. Transformer-based methods solve the above problems through parallel computing. Moreover, transformer-based methods can learn the feature with global dependence. As shown in Fig. 7(h), IBot results are greatly improved,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

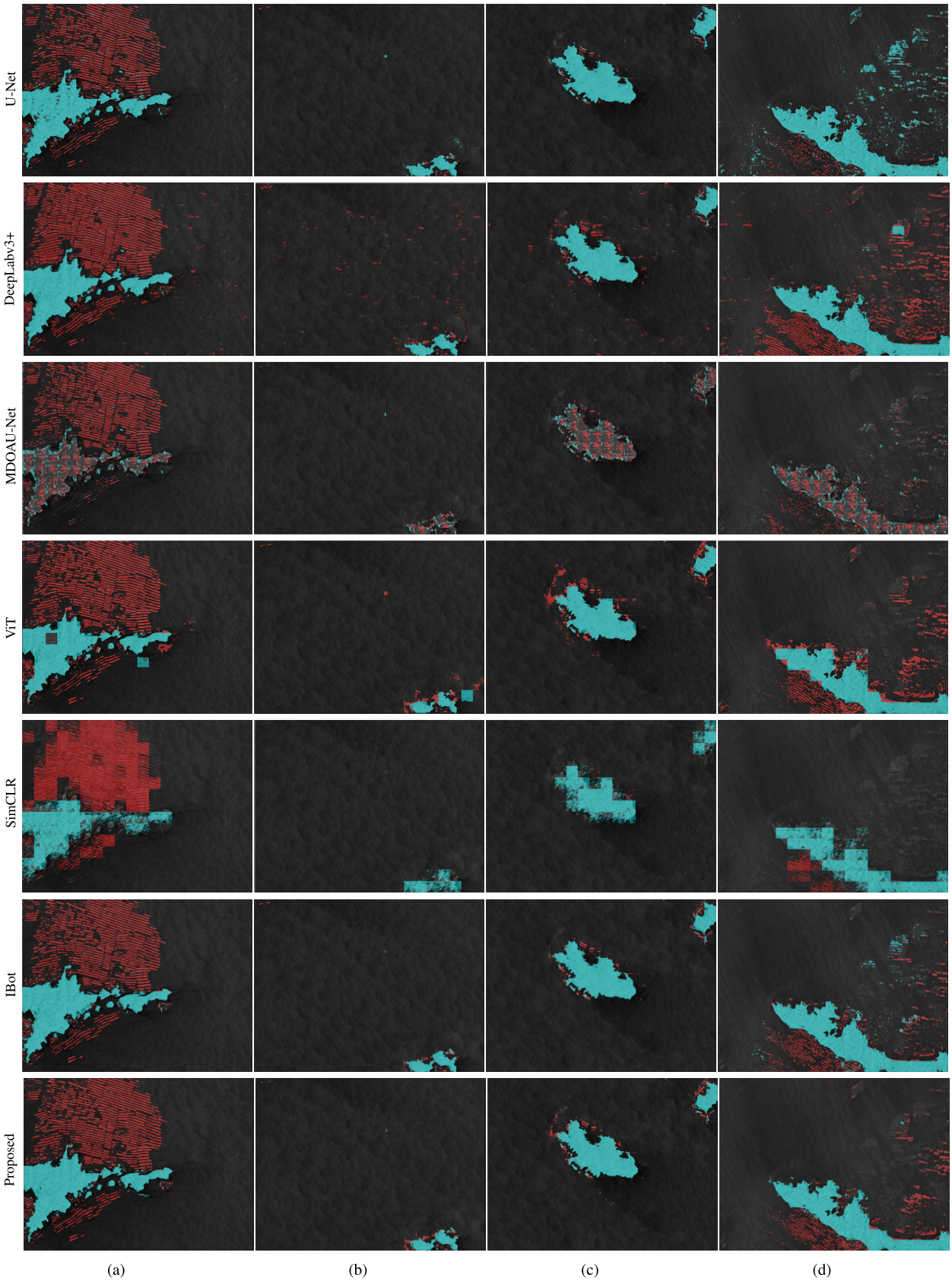


Fig. 6. Segmentation images of the GF-3 aquaculture dataset based on the whole remote-sensing images. (a) Area A. (b) Area B. (c) Area C. (d) Area D.

but there are still some misjudgments and missing detection. Although IBot considers multifeature and improves the accuracy of most image processing tasks, it does not consider deep semantic enhancement. For complex SAR images, their deep semantic features are critical. There is a lack of segmentation continuity in the aquaculture areas. As shown in Fig. 7(i),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

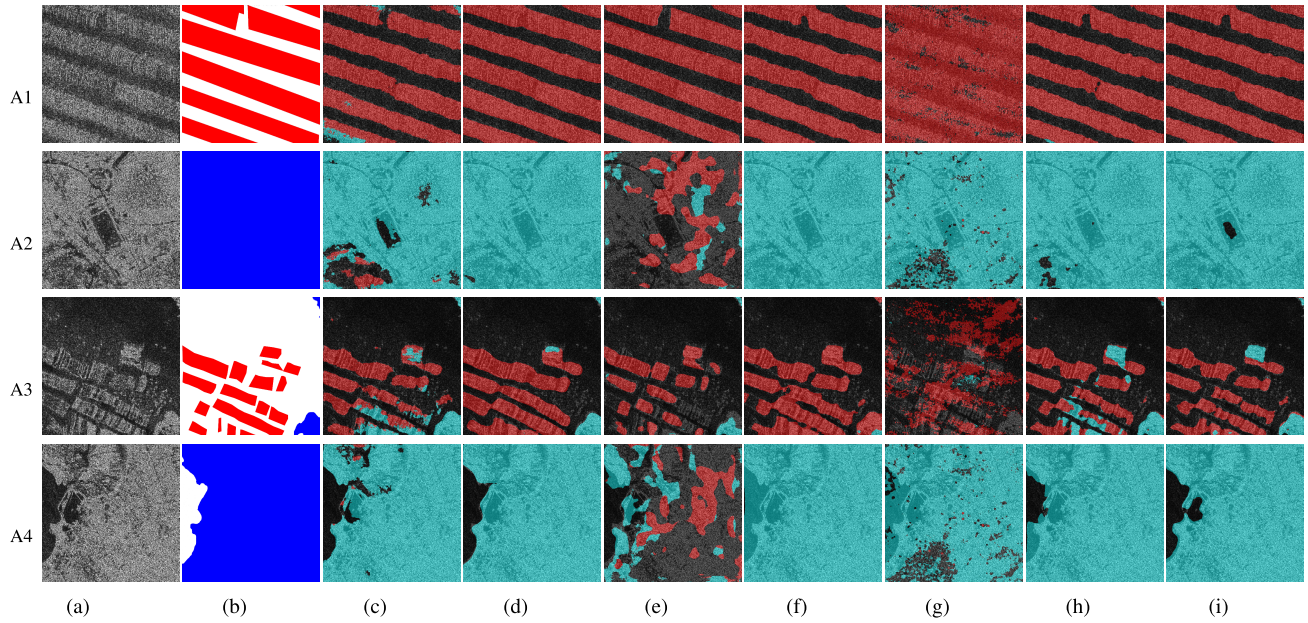


Fig. 7. Segmentation image details of GF-3 aquaculture dataset. (a) Original images. (b) Ground truth. (c) U-Net. (d) Deeplabv3+. (e) MDOAU-Net. (f) ViT. (g) SimCLR. (h) Ibot. (i) STFF.

TABLE I
RESULTS OF GF-3 AQUACULTURE DATASET

Methods	Area	*mIoU*	*Kappa*	*OA*	*P*	*R*
U-Net [10]	A	0.6179	0.5617	94.23	50.34	50.73
	B	0.8376	0.7319	99.56	86.86	86.80
	C	0.8659	0.8273	98.93	87.49	87.61
	D	0.4287	0.2336	88.95	32.33	32.46
	Mean	0.6875	0.5886	95.42	64.26	64.40
Deeplabv3+ [6]	A	0.6773	0.4846	94.39	89.60	87.32
	B	0.6389	0.3056	98.64	98.77	97.79
	C	0.6980	0.4703	97.96	91.95	93.10
	D	0.5619	0.3145	91.05	77.62	80.55
	Mean	0.6440	0.3938	95.51	**89.49**	89.69
MDOA U-Net [14]	A	0.7658	0.7586	88.62	76.73	78.04
	B	0.9686	0.9605	98.54	98.29	98.06
	C	0.8754	0.8614	93.10	90.22	89.26
	D	0.5937	0.4828	81.30	59.87	61.46
	Mean	0.8009	0.7658	90.39	81.28	81.71
ViT [12]	A	0.8133	0.7946	93.52	83.85	85.25
	B	0.9651	0.9559	98.84	97.53	97.71
	C	0.9112	0.8972	97.71	91.19	93.59
	D	0.7342	0.6259	89.76	82.23	89.70
	Mean	0.8560	0.8184	94.96	88.70	91.56
SimCLR [25]	A	0.5589	0.3882	82.62	57.86	57.54
	B	0.8932	0.8011	98.97	98.24	98.26
	C	0.8657	0.7962	96.80	89.73	90.10
	D	0.7660	0.6683	92.15	76.41	76.86
	Mean	0.7710	0.6635	92.64	80.56	80.69
IBot [31]	A	0.8122	0.8022	93.44	79.50	81.78
	B	0.9760	0.9717	99.34	98.22	98.32
	C	0.9231	0.9195	98.70	92.33	93.27
	D	0.6781	0.5855	90.34	65.89	68.18
	Mean	0.8474	0.8197	95.46	83.99	85.39
STFF	A	0.8346	0.8197	94.51	83.82	87.52
	B	0.9784	0.9761	99.35	98.38	98.89
	C	0.9140	0.9047	98.51	89.94	92.46
	D	0.7281	0.6180	89.79	80.83	89.41
	Mean	**0.8638**	**0.8296**	**95.54**	88.24	**92.07**

the proposed one includes shallow image features and enhances deep semantic features through the multilayer feature fusion module, showing the best performance.

The detailed results of the experiments are tabulated in Table I. The OA index focuses on the overall classification accuracy, and the accuracy is affected by the classification of the main categories. In whole-scene remote-sensing images, seawater accounts for a large proportion. In the case that most seawater is classified correctly, each comparison method has a good performance on OA. Often, there is an inverse relationship between *P* and *R*, where it is possible to increase one at the cost of reducing the other. The proposed method performs better in *R*, indicating that more true samples are found, i.e., more mariculture was extracted from all the true mariculture samples. This is important in monitoring tasks. Deeplabv3+ performs better in *P*, indicating that there are more true samples in the prediction results, i.e., more mariculture was correct from all the mariculture prediction samples. Although *P* of the proposed is slightly lower than Deeplabv3+, in order to monitor all illegal farming behaviors, a slightly lower *P* is acceptable. The *kappa* indicator tests the consistency of the predicted results with the ground truth. The *kappa* of the proposed one is the best, and Deeplabv3+ has a very low *kappa* due to misjudgment in most seawater areas. MDOAU-Net seems to have poor land consistency, but the land area accounts for a small area of the whole image, so *kappa* is higher than Deeplabv3+. The mIoU index is the intersection over the union of the prediction result and the ground truth, which is mainly affected by misjudgment, especially when the ground truth is one class, there are multiple classes in the prediction result, and the mIoU declines significantly. Although U-Net has much less misjudgment of seawater, due to many misjudgments in the aquaculture area and land area, the mIoU of U-Net is not much higher than that of Deeplabv3+, which is different from *kappa*. The proposed method has a good performance of the mIoU. At the same time, it reduces the misjudgment of seawater area, aquaculture area, and land area. Overall, the proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

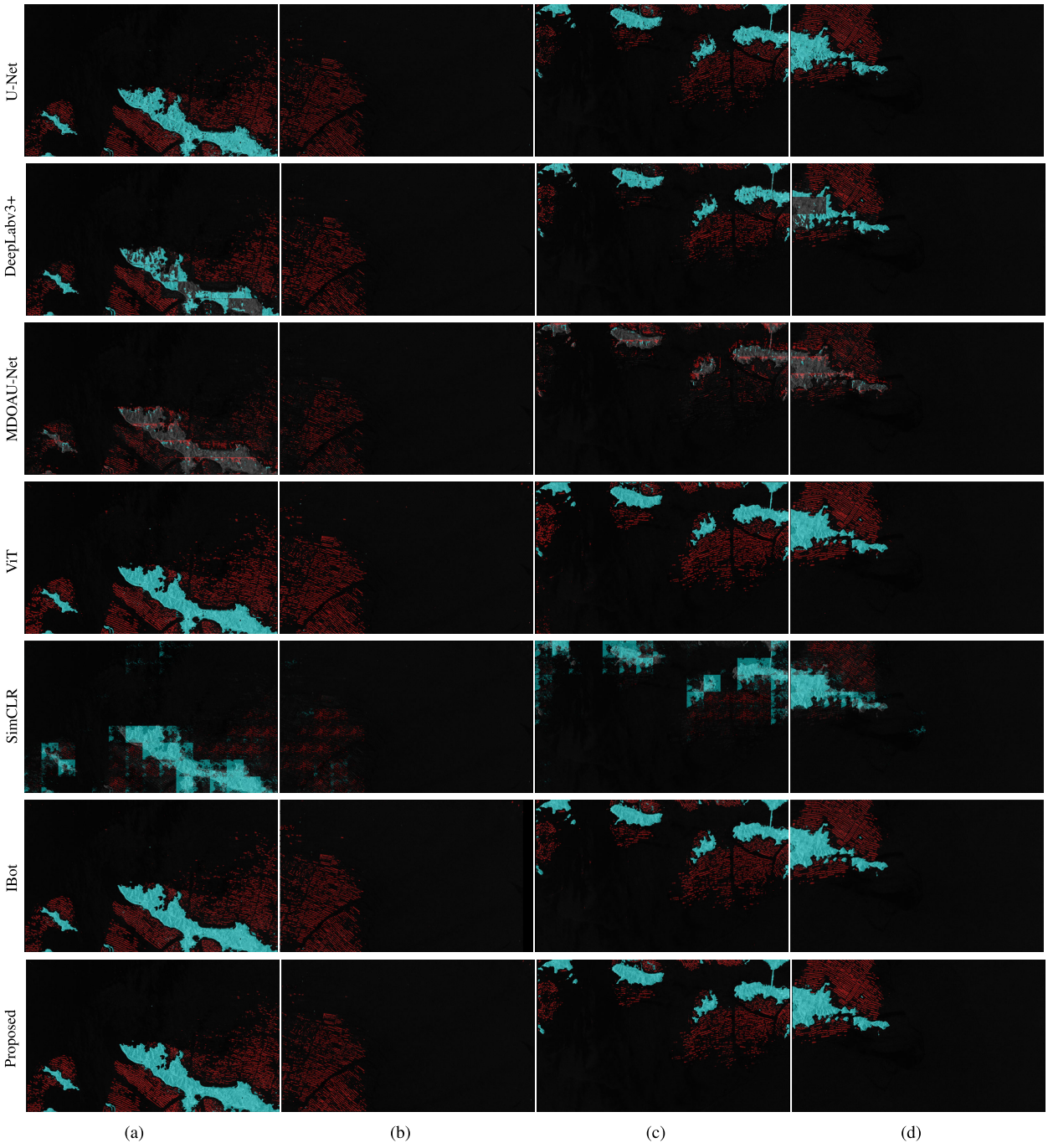


Fig. 8. Segmentation images of the Radarsat-2 aquaculture dataset based on the whole remote-sensing images. (a) Area E. (b) Area F. (c) Area G. (d) Area H.

method achieves satisfactory results in all evaluation indicators except *P*. In addition, self-supervised STFF fine-tuning training time is similar to that of Deeplabv3+ in the same epoch. In the test, the frames per second is 354 s for STFF and 64 s for Deeplabv3+.

C. Results on Radarsat-2 Marine Aquaculture Datasets

The experimental results of Radarsat-2 datasets are shown in Fig. 8. The corresponding accuracy is tabulated in Table II, which suggests that the proposed achieves the best performance on Radarsat-2 datasets. Fig. 8(a) shows that the rectangular phenomenon of SimCLR is the same as GF-3 datasets. Fig. 8(b) shows that the accuracy of MDOAU-Net decreases significantly for the low-resolution aquaculture area. Fig. 8(c) shows that ViT and IBot show sporadic aquaculture misclassification in the lower left seawater area due to coherent noises. Fig. 8(d) shows that Deeplabv3+ land area is discontinuous, U-Net land area has sporadic misclassification, and continuity is weaker. In addition, four areas of interest are displayed segmentation details of Radarsat-2 datasets, as shown in Fig. 9.

CNN-based methods have a similar problem with GF-3 datasets: insufficient semantic continuity. It also brings new

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

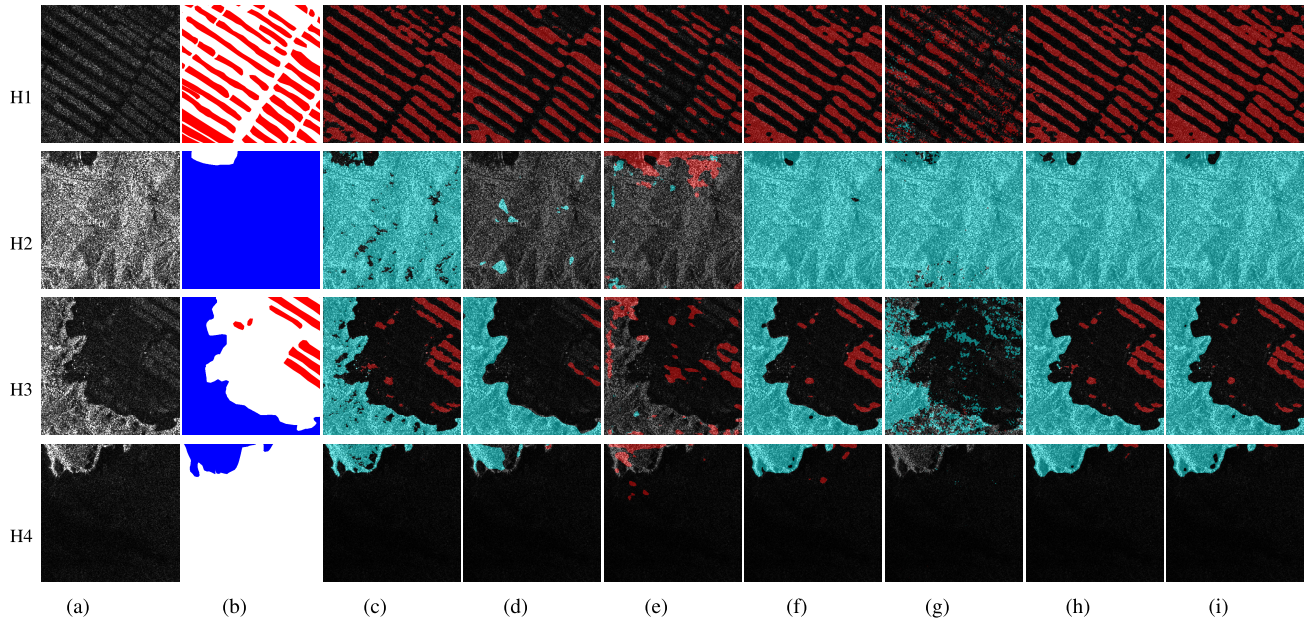


Fig. 9. Segmentation image details of Radarsat-2 aquaculture dataset. (a) Original images. (b) ground truth. (c) U-Net. (d) Deeplabv3+. (e) MDOAU-Net. (f) ViT. (g) SimCLR. (h) IBot. (i) STFF.

TABLE II
RESULTS OF RADARSAT-2 AQUACULTURE DATASET

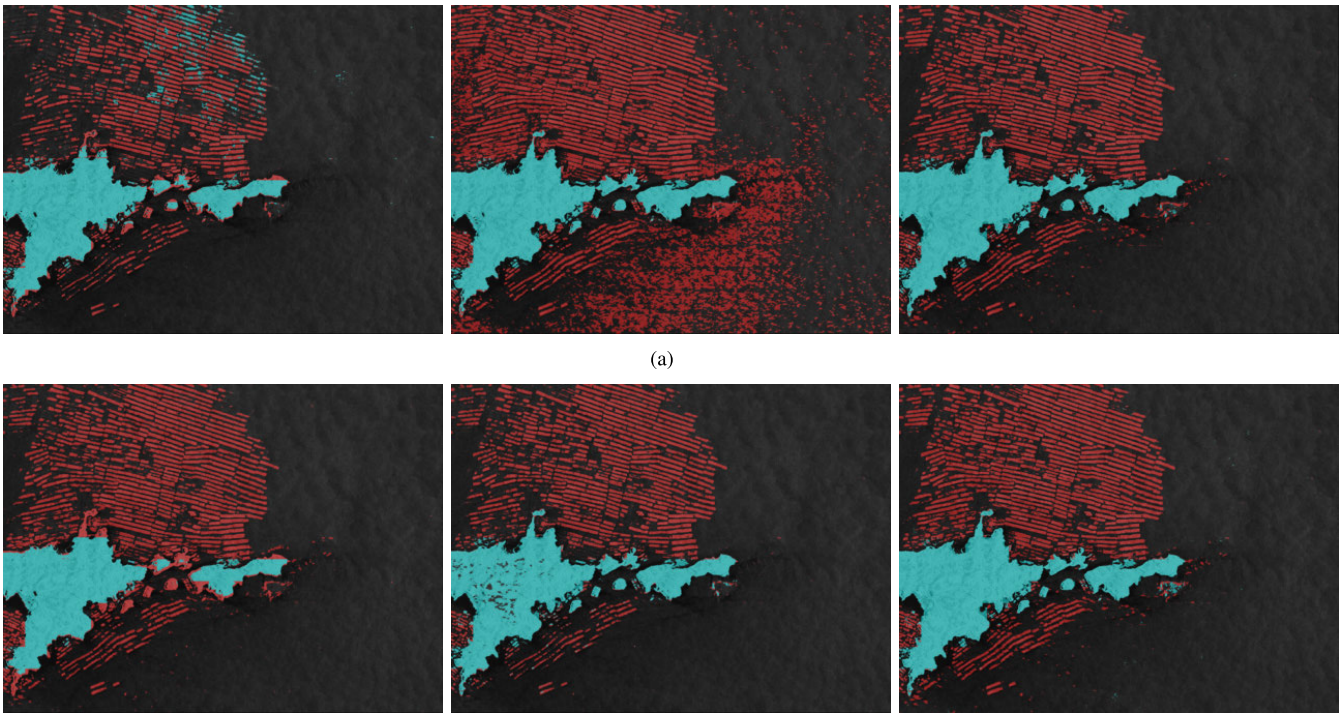
Methods	Area	*mIoU*	*Kappa*	*OA*	*P*	*R*
U-Net [10]	E	0.6988	0.6378	94.45	76.12	74.27
	F	0.8000	0.7115	97.68	83.86	84.63
	G	0.7884	0.7530	96.44	85.62	84.95
	H	0.9139	0.9161	98.33	89.72	89.19
	Mean	0.8003	0.7546	96.73	83.83	83.26
Deeplabv3+ [6]	E	0.7419	0.6911	90.88	79.21	85.00
	F	0.9036	0.8697	97.51	95.00	96.23
	G	0.7859	0.7404	95.08	84.76	87.37
	H	0.9126	0.9138	95.89	94.08	93.82
	Mean	0.8360	0.8038	94.84	88.26	90.61
MDOA U-Net [14]	E	0.6824	0.5791	85.17	80.00	83.13
	F	0.8644	0.7813	96.48	96.50	99.56
	G	0.6880	0.5564	89.31	82.51	84.44
	H	0.8645	0.8316	93.52	91.56	91.57
	Mean	0.7748	0.6871	91.12	87.64	89.68
ViT [12]	E	0.7113	0.6302	94.63	82.17	83.28
	F	0.8395	0.7447	97.76	94.16	94.27
	G	0.7690	0.7025	96.44	88.06	88.36
	H	0.9179	0.8919	98.42	94.64	94.32
	Mean	0.8094	0.7423	96.81	89.76	90.06
SimCLR [25]	E	0.5119	0.2697	87.76	56.78	57.23
	F	0.8040	0.7115	94.67	77.78	77.76
	G	0.6334	0.4766	88.65	64.98	66.11
	H	0.7998	0.7074	95.37	85.15	84.00
	Mean	0.6873	0.5413	91.61	71.17	71.28
IBot [31]	E	0.7315	0.6593	95.10	85.53	84.79
	F	0.8689	0.7958	97.84	96.27	96.74
	G	0.8188	0.7925	96.00	90.05	89.60
	H	0.9355	0.9242	98.64	95.71	94.90
	Mean	0.8387	0.7930	96.90	91.89	91.51
STFF	E	0.7766	0.7465	95.34	86.35	85.10
	F	0.9058	0.8680	97.75	96.78	97.66
	G	0.8445	0.8444	96.85	89.32	89.77
	H	0.9462	0.9493	98.48	95.86	94.26
	Mean	**0.8683**	**0.8521**	**97.11**	**92.08**	**91.70**

problems due to different satellite resolutions. The aquaculture regions in one image become more aquaculture areas and slenderer. U-Net and Deeplabv3+ have the phenomenon of information loss. That is, the extraction of continuous aquaculture regions is discontinuous due to the lack of deep semantics. As shown in Fig. 9(c), U-Net judged the dark area of land as seawater during land segmentation. The smaller resolution of Radarsat-2 will cause the size of the image with the same geographical range to be smaller than that of GF-3. The problem is that the dataset is smaller, and the land samples are few and unbalanced. As shown in Fig. 9(d), Deeplabv3+y is difficult to extract land due to the small number of land samples. As shown in Fig. 9(e), MDOAU-Net is designed for GF-3 datasets, and missed detection occurred after the aquaculture targets became smaller. As shown in Fig. 9(g), the performance of SimCLR is also unsatisfactory, the same as the GF-3 dataset.

Transformer-based methods effectively alleviate unbalanced samples through a globally dependent encoder. As shown in Fig. 9(f), ViT extracts continuous and nonmissing land areas. However, the deep semantic features of ViT are weak during decoding, resulting in a small amount of aquaculture misjudgment in the seawater. As shown in Fig. 9(h), IBot improves the above misjudgments. However, the aquaculture area width is shorter than the ground truth. As shown in Fig. 9(i), by the multilayer feature fusion and aquaculture edge loss, STFF captures the aquaculture area shape features of the lower layer, accurately extracting the aquaculture edge.

The detailed results of the experiments are tabulated in Table II. Similar to GF-3 datasets, Radarsat-2 datasets have a large proportion of seawater, so there is a high OA. Due to the imaging and the small proportion of land, the mIoU and *kappa* of Deeplabv3+ are significantly improved compared to the GF-3 dataset. The seawater in the Radarsat-2 dataset is smoother, which benefits Deeplabv3+. Similarly, U-Net also performs better on mIoU and *kappa* than the GF-3 dataset. The original intention of MDOAU-Net is to extract the aquaculture with the GF-3 dataset, so mIoU and *kappa* have declined

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.



(a)

(b)

Fig. 10. Segmentation images of different labeling proportion in area A of GF-3 satellite images. From left to right are the results of 3, 15, and 30 labels. (a) IBot results based on the whole remote-sensing image. (b) STFF results based on the whole remote-sensing image.

on the Radarsat-2 dataset. STFF reduces the occurrence of misjudgments in the seawater, ensures semantic continuity in aquaculture area and land, and achieves satisfactory evaluation in all indexes. In addition, since the image sizes of the Radarsat-2 and GF-3 datasets are the same, the training time and test frames per second of STFF and Deeplabv3+y perform similarly as on the GF-3 datasets. The frames per second of STFF is 324 s, and Deeplabv3+ is 58 s. Although STFF takes more time compared with Deeplabv3+ during testing, in remote-sensing image processing, due to the complexity of images and the huge amount of data, the processing time of STFF is acceptable.

D. Labeling Part

In this section, the number of labels is mainly discussed on the influence of STFF. Labeling production is a time-consuming task in the face of increasing remote-sensing data. Self-supervised learning obtains better features through unsupervised training and completes the downstream task with a few labels. There are 3, 15, and 30 labels to discuss the performance of self-supervised STFF and IBot under different labels, and the results are shown in Fig. 10 and Table III.

As shown in Fig. 10(a), IBot showed a large area of misjudgment in the aquaculture area in the case of three labels. It is semantically unreasonable to regard aquaculture as land because the shape characteristics and area of land differ from aquaculture. In the case of 15 labels, IBot has many sporadic aquaculture misjudgments in seawater areas. The generalization ability of seawater feature learning is not enough to cover most sea states with speckle noise. At 30 labels, IBot starts to perform satisfactorily.

TABLE III
PERFORMANCE CRITERIA OF THE LABELING PROPORTION EXPERIMENTAL RESULTS ON GF-3 DATASET

	Method.	*mIoU*	*Kappa*	*OA*	*P*	*R*
3	IBot	0.7056	**0.6853**	91.60	65.15	67.89
	STFF	**0.7248**	0.6356	**92.47**	**79.99**	**84.58**
15	IBot	0.5283	0.3164	82.90	**85.07**	75.13
	STFF	**0.7664**	**0.7180**	**93.84**	78.40	**82.45**
30	IBot	0.7260	0.6430	93.87	77.20	77.81
	STFF	**0.7680**	**0.7198**	**93.96**	**80.60**	**82.12**

As shown in Fig. 10(b), STFF enhances the semantics and alleviates the misjudgment of aquaculture as land in the case of three labels. Note that the 3, 15, and 30 labels of STFF are the same as IBot. With only one land label, one aquaculture label, and one seawater label, STFF obtained satisfactory results. With 15 labels, STFF improves the extraction of land edges. In the case of 30 labels, STFF further optimizes land continuity.

Overall, STFF performs better than IBot with the same labels, as shown in Table III. Especially on a small number of labels, STFF performs significantly better than IBot with 3 and 15 labels. When there are three labels, the mIoU of STFF is higher than IBot's. This is because IBot's seawater and aquaculture areas have been misjudged, and there are cases where the category labels do not correspond. The *kappa* of STFF is lower than that of IBot. This is because STFF has a misjudgment at the edge of the land, and the misjudgment area is larger than the misjudgment area of IBot, resulting in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
PERFORMANCE CRITERIA OF THE LABELING PROPORTION EXPERIMENTAL RESULTS ON RADARSAT-2 DATASET

	Method.	*mIoU*	*Kappa*	*OA*	*P*	*R*
3	IBot	0.7531	0.5861	96.00	93.61	92.58
	STFF	**0.7879**	**0.6481**	**97.28**	**94.50**	**94.00**
15	IBot	0.8903	0.8456	98.32	94.17	**93.46**
	STFF	**0.9126**	**0.8900**	**98.34**	**94.17**	93.41
30	IBot	**0.9436**	**0.9508**	98.57	94.04	93.55
	STFF	0.9424	0.9435	**98.58**	**94.80**	**94.24**

TABLE V
RESULTS OF THE ABLATION EXPERIMENTS

Edge Loss Function	Feature Fusion Module	*MIoU*	*OA*(%)
		0.8474	95.46
✓		0.8625	95.54
	✓	0.8628	95.55
✓	✓	0.8638	95.54

a lower *kappa* of STFF. But this is acceptable because it is reasonable to prioritize ensuring aquaculture's extraction accuracy in mariculture monitoring. The accuracy of STFF is close to that of IBot under 30 labels.

The results of the Radarsat-2 dataset are tabulated in Table IV. Similar to GF-3, STFF performs significantly better than IBot with 3 and 15 labels. The difference in accuracy between STFF and IBot on the 30 labels is not significant.

E. Ablation Part

This section will discuss the effect of the effectiveness of the edge loss function and the feature fusion module. Table V shows that the edge loss function and the feature fusion module have effectively improved. Enhancing the feature fusion module of deep semantics and the edge loss function of edge information can improve intraclass continuity and reduce interclass misjudgment from different perspectives. They all have different degrees of improvement to the model and a small degree of improvement after the merger.

In addition, the OA is mainly affected by the overall classification accuracy of images, and the mIoU focuses on detailed semantics. In Table V, the OA improvement is insignificant, indicating that the proposed method has an effective recognition ability for seawater targets in all settings of the ablation experiment. The significant improvement of mIoU indicates that the proposed method acquires semantic information, avoids misclassification, and enhances continuity. In detail, the object of OA evaluation is mainly the image's seawater portion. If most seawater is correctly classified, it will not cause a significant change in OA. The object of mIoU evaluation is semantic information, and the existence of aquaculture on the far sea is not reasonable from the semantic point of view. The prediction changes from the one-class classification of only seawater to two-class classifications where seawater and aquaculture coexist, resulting in a rapid decline in mIoU. The OA gives an overall accuracy but not the semantics of the results. The mIoU is more reasonable in evaluating the results of the whole-scene image. It illustrates one of the particularities of the whole-scene marine aquaculture extraction problem. The proposed method improves the semantic information of the results while ensuring overall classification accuracy.

V. CONCLUSION

This article presents an STFF for SAR marine aquaculture segmentation. The encoder of STFF is based on the transformer, utilizing the hybrid loss function to obtain the features with global, local, and edge information. The decoder of STFF is to strengthen the deep semantic information and fuse it with multilayer features while improving precision and continuity. STFF can obtain explicit features even in multitarget sample imbalance and reduce the dependence on labels by extracting features from many unlabeled data. STFF can accurately extract multiple targets with different resolutions. Based on the idea of enlarging interclass differences and narrowing intraclass differences, STFF avoids target misjudgment and enhances target continuity. Experimental results illustrate that the proposed method can handle whole-scene remote-sensing images with superior results. With high resolution, the proposed method improves the phenomenon of misclassification of aquaculture areas, enhancing the semantic continuity of the extraction results. With low resolution, the proposed method increases the extraction precision of aquaculture floating raft edges. In addition, STFF is suitable for a small number of labels and multiple-target segmentation issues because of its ability to acquire rich semantic features. However, targets are diverse at sea, and although STFF segments out the primary targets, there are also tiny targets, such as fishing boats docked near the shore, interfering with the accuracy improvement. Moreover, sea conditions with strong wind and waves result in significant differences in the imaging characteristics of mariculture, interfering with the extraction accuracy. The modification of tiny targets and weak targets recognition ability is urgently required in the future.

REFERENCES

[1] R. L. Naylor et al., "A 20-year retrospective review of global aquaculture," *Nature*, vol. 591, no. 7851, pp. 551–563, Mar. 2021.

[2] R. R. Gentry et al., "Mapping the global potential for marine aquaculture," *Nature Ecol. Evol.*, vol. 1, no. 9, pp. 1317–1324, Aug. 2017.

[3] Q. Xing et al., "Monitoring seaweed aquaculture in the Yellow Sea with multiple sensors for managing the disaster of macroalgal blooms," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111279.

[4] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

[5] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[6] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 801–818.

[7] H. Du, J. Wang, M. Liu, Y. Wang, and E. Meijering, "SwinPA-Net: Swin transformer-based multiscale feature pyramid aggregation network for medical image segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Sep. 19, 2022, doi: 10.1109/TNNLS.2022.3204090.

[8] Z. Li et al., "HCRB-MSAN: Horizontally connected residual blocks-based multiscale attention network for semantic segmentation of buildings in HSR remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5534–5544, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.

[9] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Central attention network for hyperspectral imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 10, 2022, doi: 10.1109/TNNLS.2022.3155114.

[10] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention*. Munich, Germany: Springer, 2015, pp. 234–241.

[11] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[12] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–22.

[13] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023.

[14] J. Wang, J. Fan, and J. Wang, "MDOAU-Net: A lightweight and robust deep learning model for SAR image segmentation in aquaculture raft monitoring," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 4504505, doi: 10.1109/LGRS.2022.3147355.

[15] T. Shi, Q. Xu, Z. Zou, and Z. Shi, "Automatic raft labeling for remote sensing images via dual-scale homogeneous convolutional neural network," *Remote Sens.*, vol. 10, no. 7, p. 1130, Jul. 2018.

[16] B. Cui, D. Fei, G. Shao, Y. Lu, and J. Chu, "Extracting raft aquaculture areas from remote sensing images via an improved U-Net with a PSE structure," *Remote Sens.*, vol. 11, no. 17, p. 2053, Sep. 2019.

[17] B. Cheng, C. Liang, X. Liu, Y. Liu, X. Ma, and G. Wang, "Research on a novel extraction method using deep learning based on GF-2 images for aquaculture areas," *Int. J. Remote Sens.*, vol. 41, no. 9, pp. 3575–3591, May 2020.

[18] J. Fan, J. Zhao, W. An, and Y. Hu, "Marine floating raft aquaculture detection of GF-3 PolSAR images based on collective multikernel fuzzy clustering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 2741–2754, Aug. 2019.

[19] Y. Hu, J. Fan, and J. Wang, "Target recognition of floating raft aquaculture in SAR image based on statistical region merging," in *Proc. 7th Int. Conf. Inf. Sci. Technol. (ICIST)*, Apr. 2017, pp. 429–432.

[20] X. Wang, J. Zhou, and J. Fan, "IDUDL: Incremental double unsupervised deep learning model for marine aquaculture SAR images segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4209412, doi: 10.1109/TGRS.2022.3203071.

[21] L. Jing and Y. Tian, "Self-supervised visual feature learning with deep neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 11, pp. 4037–4058, Nov. 2021.

[22] X. Liu et al., "Self-supervised learning: Generative or contrastive," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 857–876, Jan. 2023.

[23] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, Jul. 2006.

[24] I. Goodfellow et al., "Generative adversarial networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2014, no. 11, pp. 1–9.

[25] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[26] J. Kang, R. Fernandez-Beltran, P. Duan, S. Liu, and A. J. Plaza, "Deep unsupervised embedding for remotely sensed images based on spatially augmented momentum contrast," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2598–2610, Mar. 2021.

[27] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

[28] M. Caron, I. Misra, J. Mairal, P. Goyal, P. Bojanowski, and A. Joulin, "Unsupervised learning of visual features by contrasting cluster assignments," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 9912–9924.

[29] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6000–6010.

[30] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9630–9640.

[31] J. Zhou et al., "IBot: Image BERT pre-training with online tokenizer," in *Proc. ICLR*, 2021, pp. 1–12.

[32] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jul. 15, 2022, doi: 10.1109/TNNLS.2022.3189994.

[33] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410313, doi: 10.1109/TGRS.2022.3168697.

[34] H.-F. Zhong, Q. Sun, H.-M. Sun, and R.-S. Jia, "NT-Net: A semantic segmentation network for extracting lake water bodies from optical remote sensing images based on transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627513, doi: 10.1109/TGRS.2022.3197402.

[35] X. Meng, Y. Yang, L. Wang, T. Wang, R. Li, and C. Zhang, "Class-guided Swin transformer for semantic segmentation of remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6517505, doi: 10.1109/LGRS.2022.3215200.

[36] L. Ma, J. Li, H. Guan, Y. Yu, and Y. Chen, "STN: Saliency-guided transformer network for point-wise semantic segmentation of urban scenes," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 7004405, doi: 10.1109/LGRS.2022.3190558.

[37] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[38] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[39] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149.

[40] A. Raganato and J. Tiedemann, "An analysis of encoder representations in transformer-based machine translation," in *Proc. EMNLP Workshop BlackboxNLP, Analyzing Interpreting Neural Netw. NLP 1st Workshop*, 2018, pp. 287–297.

[41] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715, doi: 10.1109/TGRS.2022.3144165.

[42] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820, doi: 10.1109/TGRS.2022.3144894.

[43] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.



Jianchao Fan (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Control Science and Engineering, Dalian University of Technology, Dalian, China, in 2007 and 2012, respectively.

He was a Post-Doctoral Fellow jointly cultivated by The Chinese University of Hong Kong, Hong Kong, and the Dalian University of Technology from 2013 to 2016 and also a Visiting Scholar with the Department of Computer Science and Engineering, Washington University in St. Louis, St. Louis, MO, USA, from 2018 to 2019. He is currently a Professor with the School of Control Science and Engineering, Dalian University of Technology. He has been a Professor with the Department of Ocean Remote Sensing, National Marine Environmental Monitoring Center, Dalian, since 2019. He has authored/coauthored three books and over 90 scientific papers in international journals and conference proceedings, such as IEEE TRANSACTIONS ON FUZZY SYSTEMS, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Pattern Recognition*. His research interests include marine SAR image processing, polarimetric SAR, deep learning neural networks, and heuristic swarm intelligent optimization.

Dr. Fan serves as the Program Chair, the Workshop Chair, the Publicity Chair, and the Local Arrangement Chair for several international conferences, such as the International Symposium on Neural Networks, the International Conference on Information Society and Technology, and the International Conference on Intelligent Control and Information Processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.



Jianlin Zhou received the B.S. and M.S. degrees from Dalian Polytechnic University, Dalian, China, in 2020 and 2023, respectively. He is currently pursuing the Ph.D. degree with the School of Automation, Central South University, Changsha, China.

His main research interests include unsupervised deep learning and remote sensing image processing.



Xinzhe Wang received the Ph.D. degree from the School of Control Science and Engineering, Dalian University of Technology, Dalian, China, in 2012.

He is currently with the School of Information Science and Engineering, Dalian Polytechnic University, Dalian. His research interests include data-driven modeling and deep learning neural networks.



Jun Wang (Life Fellow, IEEE) received the B.S. degree in electrical engineering and the M.S. degree in systems engineering from the Dalian University of Technology, Dalian, China, in 1982 and 1985, respectively, and the Ph.D. degree in systems engineering from Case Western Reserve University, Cleveland, OH, USA, in 1991.

He is currently the Chair Professor with the Department of Computer Science, School of Data Science, City University of Hong Kong, Hong Kong. Prior to this position, he held various academic positions at the Dalian University of Technology; Case Western Reserve University; the University of North Dakota, Grand Forks, ND, USA; and the Chinese University of Hong Kong. He also held various part-time visiting positions at the USAF Armstrong Laboratory, San Antonio, TX, USA; Japanese Brain Science Institute, Wako, Japan; Huazhong University of Science and Technology, Wuhan, China; Dalian University of Technology; and Shanghai Jiao Tong University, Shanghai, China.

Dr. Wang is a fellow of the International Association for Pattern Recognition and a Foreign Member of Academia Europaea (Academy of Europe). He was a recipient of the Research Excellence Award from The Chinese University of Hong Kong, Hong Kong, from 2008 to 2009, the Outstanding Achievement Award from Asia Pacific Neural Network Assembly, IEEE TRANSACTIONS ON NEURAL NETWORKS Outstanding Paper Award in 2011, the Neural Networks Pioneer Award from the IEEE Computational Intelligence Society in 2014, the AI Science and Technology Achievement Award from the Chinese Association for Artificial Intelligence in 2016, and the Norbert Wiener Award from the IEEE Systems, Man and Cybernetics Society in 2019, among other distinctions. He served as the General Chair of the 13th/25th International Conference on Neural Information Processing in 2006 and 2018 and the IEEE World Congress on Computational Intelligence in 2008, and the Program Chair of the IEEE International Conference on Systems, Man, and Cybernetics in 2012. He was an Editor-in-Chief of IEEE TRANSACTIONS ON CYBERNETICS from 2014 to 2019 and an Associate Editor of the journal and its predecessor from 2003 to 2013. He also served as an Associate Editor for IEEE TRANSACTIONS ON NEURAL NETWORKS and IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS—PART C: APPLICATIONS AND REVIEWS and a member of an Editorial Advisory Board of *International Journal of Neural Systems* and an Editorial Board of *Neural Networks*. He was an IEEE Computational Intelligence Society Distinguished Lecturer from 2010 to 2012 and from 2014 to 2016, and an IEEE Systems, Man, and Cybernetics Society Distinguished Lecturer from 2017 to 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:12:12 UTC from IEEE Xplore. Restrictions apply.